

Field-deployable inputs: low-cost LES, K sparse sensors, and the PDE -- no DNS full field in training.

DNS full field
offline evaluation only

▼ compare

Low-cost LES

no DNS
statistical surrogate

POD + QR-pivot

K = 100 sensors

QR-pivot placement
measure u, v in time

sensor time series

PI-CON operator

sensor MSE + PDE residual
(no full-field target)

query (x, y, t)

Reconstructed field

u, v, p over full domain

phase-decorrelated;
places sensors only

actual realisation,
energy-dominant scales

The LES proxy places the sensors but cannot reconstruct the realisation; PI-CON does, from the placement and sensor histories alone.